DR AMOL DESHMUKH

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RESEARCH EXPERIENCE

University of Glasgow, Glasgow, UK *April 2016-current*
Research Associate

- Developed non-verbal behaviour, gestures for virtual agents using **computational modelling** working with Prof. Stacy Marsella at the School of Psychology (since Aug. 2019 - current).

MuMMER Project - School of Computing science - EU Horizon 2020 *April 2016 - July 2019*

- Project Goal: Develop a humanoid robot that can interact autonomously and naturally in the dynamic environments of a shopping mall, providing an engaging and entertaining experience to users. (Supervisor Dr Mary Ellen Foster)
- Developed social **signal processing** components to convey personality dimensions using parameterised gestures, multi-modal non-verbal behaviour generation for a shopping mall robot (Pepper).
- Implemented machine learning algorithms to recognise engagement/disengagement of users, infer users' social state interest and attention, user studies to **analyse** user's perception on gestures.

Heriot-Watt University, Edinburgh, UK *September 2008 - March 2016*
Research Associate - School of Mathematical & Computer Sciences - Supervisor Prof. Ruth Aylett

- EMOTE (December 2012 - March 2016): The project goal was to develop empathic robotic tutors for children in schools.
  Role and Responsibilities:
  - **Evaluated** interaction effects between children and a robotic tutor, on appearance, expressiveness and memory in a pedagogical setting.
  - Integrated **theoretical and pedagogical** work by project partners using **user-centred design** approach.
  - Integration software modules (vision and perception), interaction modalities into a generic architectural framework.
  - **Led technical meetings** with project team members, produced test plans.
  - Conducted user studies with School children in Scotland.
- JAMES (September 2012-November 2012): The project explored the core cognitive capabilities that are required for a robot bartender to interact with humans in a socially-appropriate manner.
  - Developed interaction behaviours for NAO humanoid robot, analysed interaction video data.
- LIREC, EU FP7 (September 2008-August 2012): The project aimed to build a new generation of interactive, emotionally intelligent companion robots, capable of establishing long-term relationship with humans in real social environments like homes and offices.
  Role and Responsibilities:
  - **Investigated** the service degradation caused by recharging and explored social verbal strategies used to influence users expectations during recharge, **long-term deployment**.
  - Integrated system modules, developed competencies for robot e.g. human-aware navigation, user monitoring, vision perception, autonomous recharging.
  - Designed and engineered hardware (electronics, mechanical) for mobile social robot.
  - Developed **technical capabilities** for social mobile robot, work-package deliverables for EU.
  - Collaborate with project partners in the project, led technical meetings with project partners.
- Designed and constructed the mechanical and electrical components of social robot platform.
- Supervised 7 M.Sc projects, demo presentations for University open days and public events.

INDUSTRY EXPERIENCE

Bleetech Innovations, Mumbai, India *January 2018 - June 2018*
Mentor/Technical advisor

- Provided **technical advise** and recommended user-centred design approaches for their innovative Blee Watch, a smart watch designed for the needs of the deaf people.

Fraunhofer IPA, Neobotix, GPS Gmbh, Stuttgart, Germany Feb-July 2008 & June-Aug 2007
Robotics Engineer (Internship & Masters Project)

- Thesis: "An Intelligent Human Robot Interaction System for Entertainment Mobile Robot."
- Developed a system for an entertainment mobile robot used in public places such as museums, shopping malls and exhibitions (CeBIT, Hannover, Germany).
- System Architecture: **Behaviour based framework**, an intelligent mechanism that accomplishes a particular task using **Reinforcement learning**.
- Monitor the performance of behaviours during interactions with humans.
- Implemented face detection, **person tracking**, human-aware navigation, behaviour based control, monitoring and adaptations of interactions with users.
- Mobile camera robot control programming, KUKA arm control programming, workspace designing, GUI development, Mobile platform Control (C/C++, Java).

Cosmic Technologies, Pune, India January 2006 - June 2006
Software Engineer (Internship & Master Thesis)

- Online Toll Free Numbers Shopping & Management System (A.S.P/ I.I.S).

Leaps & Bounds, Pune, India June 2002 - January 2003
Software Engineer: **Single-handedly** developed 3 web portals, full software development

EDUCATION

Heriot-Watt University, United Kingdom *September 2008 - December 2015*
Ph.D, "Human-Robot Interaction" Supervisor Prof. Ruth Aylett

- Thesis: "An Investigation Of Service Degradation In Long-term Human-robot Interaction With a Particular Reference To Recharge Behaviour." Investigated service degradation in long-term interaction due to recharge behaviour of autonomous social mobile robots and proposed an approach to manage service degradation due to recharging behaviour of the robot using **verbal strategies**.
- Technical Features: Human-aware navigation, path planning, vision perception, **user monitoring and tracking**, autonomous recharging, power management.

Jacobs University Bremen, Germany *August 2006 - August 2008*
M.Sc. Computer Science, "Smart Systems" (**Scholarship Award**, GPA 1.00, A+)

- Courses: Advanced robotics, Motion planning, Autonomous systems, Speech signal processing, **Computer vision and perception**.
- Thesis: "An Intelligent Human Robot Interaction System for Entertainment Mobile Robot".

Pune University, India *July 2003 - July 2006*
Master of Computer Application (M.C.A)

- Courses: Software Engineering, Network Programming, Artificial intelligence, Database Management, Object Oriented Programming C++/ Java, Statistical Computing
- Thesis: "Online Toll Free Numbers Shopping & Management System" (A.S.P/I.I.S).

Pune University, India *July 2000 - July 2003*
Bachelor of Science, Electronics, (B.Sc.)

- Thesis: "Gyro the Humanoid robot: A self conceptualised prototype of a humanoid robot."

RESEARCH FUNDING

- Travel grant of £3,500 from SICSA - Scottish Informatics and Computer Science Alliance for postdoctoral and Early Career Researcher Exchange to lead a project with a team from Amrita University in India, November 2017. (Independently led project under leave of absence).
- Contributed in defining and procuring robotics lab equipment for ROBOTARIUM, Heriot-Watt University, Edinburgh Centre for Robotics (https://www.edinburgh-robotics.org/), 2014.

PRIZES

- **First prize** at IEEE RO-MAN's Robotics Competition 2019, *"Handwashing robot"*.
- **Best one page statement** award on "What future research is required for personal robotics and why?", Towards Autonomous Robotic Systems Conference, Ulster, 2009.
- Master's project "Gyro the humanoid robot" - 3rd Prize at Techkriti at I.I.T Kanpur (**Asia's Largest** Model Presentation contest), 2nd Prize at Innovations, 2nd Prize at Robotsav, got immense media publicity in India, 2005.

ORGANISATION AND MEMBERSHIPS

- **Special session chair** for IEEE International Symposium on Robot and Human Interactive Communication (RO-MAN 2019), New Delhi, India, **Finance chair** (RO-MAN 2014).
- **Workshop chair** for inaugural workshop on "Social robots and artificial agents for developing countries: Challenges and opportunities" (RO-MAN 2019).
- **Editorial board member** of open access Journal of Future Robot Life (JFRL). Reviewer for conferences HRI, RO-MAN, ICSR and Journal of behavioural robotics.
- Affiliate member of Sustainable Futures in Africa (SFA) Network, Interdisciplinary network for researchers to build understanding, research, and practice in sustainability in Africa.
- Junior staff representative in the School's Research Committee. Organised events specially for RAs within the school to encourage networking, collaboration, and social activities.

INDEPENDENT RESEARCH EXPEDITIONS

- **Led** a researcher team from Amrita University and developed a proof-of-concept a social robot to influence handwashing behaviour in a rural village school with 100 children, March 2019, **BBC**.
- **Led and Pioneered** first human-robot interaction study in a developing country to study user perception and technology acceptance of a water hauling social robot in a rural village in India, November 2017.

DISSEMINATION ACTIVITIES

- Independently led research on human-robot interaction in rural India, multiple media articles globally. **IEEE Spectrum**, **BBC Click**, **The Telegraph**, Der Spiegel, March 2018.
- **Speaker** at Pint of science, "Robots, it makes sense!", Glasgow, 2017.
- Exhibitor at EXPLORATHON European Researchers' Night, "Pepper the robot", 2016.
- **Speaker** at Nesta event, London "Humanising the robot society", June 2014,
- The Scottish Informatics and Computer Science Alliance (SICSA), Demofest, November 2014.
- Edinburgh International Science Festival, "Let's Be Friends: Robot Companions", April 2012.
- CeBit, Hannover, Germany, "Exploring and Designing our future robot Companions", 2012.
- Featured in "Facial perception documentary", Royal Society Summer Exhibition, June 2011.
- Featured on Euronews (reach across 160 countries) "Futuris" Social robotics, 2010.

REFERENCES

- Prof. Stacy Marsella, Stacy.Marsella@glasgow.ac.uk, University of Glasgow, UK
- Dr Mary Ellen Foster, MaryEllen.Foster@glasgow.ac.uk, University of Glasgow, UK
- Prof. Ruth Aylett, Aylett, R.S.Aylett@hw.ac.uk, Heriot-Watt University, Edinburgh, UK.

SELECTED PUBLICATIONS

46 peer-reviewed publications (**21 as lead author**). Google Scholar (Citations 231, h-index 9, i10-index 8), Full publications on: https://www.psy.gla.ac.uk/~amold/#publications

1. **Amol Deshmukh** et al., 2019, October, Influencing Hand-washing Behaviour With a Social Robot: HRI Study With School Children in Rural India, *The 28th IEEE International Conference on Robot and Human Interactive Communication (RO-MAN) (pp. 1-6)* [Independently led research], BBC news: https://tinyurl.com/us4kmbb, **IEEE Spectrum**

2. Unnikrishnan R, **Amol Deshmukh** et al., October, 2019, Design and Perception of a Social Robot to Promote Hand Washing among Children in a Rural Indian School, *The 28th IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)*

3. **Amol Deshmukh** et al., 2018, December. Shaping Robot Gestures to Shape Users' Perception: The Effect of Amplitude and Speed on Godspeed Ratings. In *Proceedings of the 6th International Conference on Human-Agent Interaction* (pp. 293-300). ACM.

4. **Amol Deshmukh** et al., 2018, August. Technology Acceptance, Sociocultural Influence and Gender Perception of Robots: A Human Robot Interaction Study with Naive Users in Rural India. In *27th IEEE International Symposium on Robot and Human Interactive Communication (RO-MAN)* (pp. 1-6). IEEE. [Independently led research] (news covered in IEEE Spectrum https://tinyurl.com/ya4mwp5l, BBC Click https://tinyurl.com/y53fr599.

5. **Amol Deshmukh** et al., 2018, August. The more I understand it, the less I like it: The relationship between understandability and Godspeed scores for robotic gestures. In 2018 *27th IEEE International Symposium on Robot and Human Interactive Communication (RO-MAN)* (pp. 216-221). IEEE.

6. **Amol Deshmukh** et al., 2018. Social impact of recharging activity in long-term HRI and Verbal strategies to Manage User expectations During recharge. *Journal of Frontiers in Robotics and AI, Journal 5*, p.23.

7. **Amol Deshmukh** et al., 2017, November. Modulating the non-verbal social signals of a humanoid robot. In *Proceedings of the 19th ACM International Conference on Multimodal Interaction* (pp. 508-509). ACM.

8. **Amol Deshmukh** et al., 2017, March. Perceptual Control Theory for Engagement and Disengagement of Users in Public Spaces. In *Proceedings of the Companion of the ACM/IEEE International Conference on Human-Robot Interaction* (pp. 105-106). ACM.

9. **Amol Deshmukh** et al., 2016, March. How expressiveness of a robotic tutor is perceived by children in a learning environment. In *11th ACM/IEEE International Conference on Human-Robot Interaction (HRI)* (pp. 423-424). IEEE.

10. **Amol Deshmukh** et al., 2015, May. An empathic robotic tutor in a map application. In *Proceedings of the 2015 International Conference on Autonomous Agents and Multiagent Systems* (pp. 1923-1924). International Foundation for Autonomous Agents and Multiagent Systems.

11. Foster, M.E., **Amol Deshmukh** et al., 2015, October. How does affective robot feedback influence learner experience in a real-world treasure hunt?. *In International Conference on Social Robotics (ICSR).*

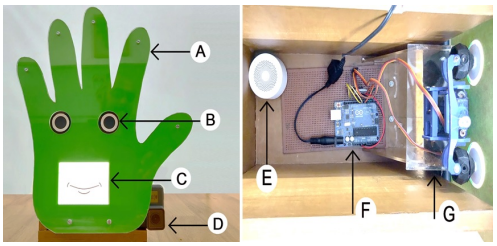
12. Lohan, K., **Amol Deshmukh** et al., 2014, August. How can a robot signal its incapability to perform a certain task to humans in an acceptable manner?. In *The 23rd IEEE International Symposium on Robot and Human Interactive Communication* (pp. 814-819). IEEE.

Social Robots as Agents for Positive Behaviour Change for Developing Countries: Good Health and Well Being

• **Background:** This fellowship aims to develop social robots as agents for **persuasive technologies** that can autonomously interact with people and effect positive change on their behaviour. Currently, social robotics research predominantly takes place in the developed world. Also, real world deployment of social robots is **rare**, with more than 75% of research done in a lab without considering real world impact.[1] In the developed world people build expectations about robots due to their exposure to media and their interaction with modern technology. On the other hand, rural populations in developing countries have limited exposure and access to different technologies due to their geographically remote and reduced economic background which make up almost half the world's population. The influence and perception of social robots on these communities has **barely** been investigated, which leaves a big **scientific gap** in the field of social robotics. In order to bridge this gap, I have identified hand hygiene as an initial case where social robots could create a very high impact in developing countries. In these countries diarrhoea and respiratory infections are the two biggest causes of child mortality, over half million young children dying a year (UNICEF). By developing an intervention such as Hand Washing With Soap (HWWS), child mortality can be reduced by 40%.[2]

• **Project Rationale:** Existing practices for influencing handwashing behaviour face three major challenges: **(i)** they are very resource intensive, **(ii)** complex to measure/monitor compliance at scale, and **(iii)** they rarely are sustainable over long-term.[3] Hence there is need for an **interdisciplinary and innovative approach** for influencing sustainable handwashing behaviour. However, **none** of the previous hand hygiene interventions have considered the use of interactive technology such as social robots. There is strong scientific evidence that the physical presence of a robot impacts human perception and influences behaviour in comparison to a screen-based agent.[4] Social robots can, for example, encourage, remind, educate, give real-time feedback, observe and monitor hand washing behaviour. Previous research in developed countries has shown that robots in education could motivate children to learn better.[5] Handwashing education in schools is an effective way of reaching children and teaching them the habit of handwashing at a young age.[6] The project is also **very timely** with recent outbreaks of COVID-19, handwashing is considered as the most effective measures to prevent spread of the virus.

• **Proof-of-concept:** Given the ambition and the risks in this research vision, I led an independent research pilot project in collaboration with Amrita University in India. A low-cost (≈ US$ 250) social robot was developed and remotely controlled (not autonomous) in a rural village school in India (see fig. 1).



A: Robot Face,
B: Eye movement,
C: Graphical screen,
D: Front camera,
E: Speaker,
F: Microcontroller,
G: Eye Mechanism

Figure 1: Image Left: Prototype Robot Design; Image Right: The wall mounted social robot near the water tap encouraged around 100 children to wash their hands via a poster on the wall, March 2019.

The short two day pilot intervention showed encouraging results: a staggering **40% increase** in levels of handwashing and with a better handwashing technique,[7] with instruction based verbal messages and eye movement from the robot[1]. However, **further research** is needed to establish a solid empirical foundation to inform interaction strategies and to develop the autonomous technology required for the social robot to influence the target subjects to perform handwashing behaviour.

• **Related Work:** In developing countries hand hygiene interventions mostly focus on educating people about health, germs and disease often using tools such as games, radio/TV campaigns, books, songs, posters, leaflets and charts.[8] Such methods can be very resource intensive and cumbersome to monitor at

[1] https://www.bbc.co.uk/news/av/technology-51360642/could-a-hand-washing-robot-help-save-lives

scale, measurements are unreliable due to self-reporting, and these approaches **rarely** result in positive, sustained behaviour change.[9] In developed countries, hand hygiene interventions have investigated the use of electronic message boards,[10] visual[11] and audio[12] prompts as a reminder to perform hand hygiene,[13] emoticon-based feedback,[14] and vision-based system for tracking people's activity in hospitals[15] with some degree of success. The state-of-the-art technology developed by SureWash (www.surewash.com) uses computer vision technology for hand hygiene training to hospital staff. SureWash costs about £30k, is bulky, works only in indoor environments and has **limited** interaction capabilities. Summing up, most technology based hand hygiene interventions are expensive, they have been performed in developed countries in indoor environments and **none** have explored the use of social robots. In regards to social robots as persuasive applications, use of verbal[16] and non-verbal behaviour such as eye gaze[17] (i.e. looking at a participants' face), has shown positive results. However, whether the persuasive effects of the robot's verbal and non-verbal cues will influence decision making in real-world scenario such as handwashing is still **undiscovered**.

- **Novelty:** Existing methods for hand hygiene promotion is carried out by, hygiene education, community-based working, incentives or advocacy, which **rarely** result in sustaining behaviour change.[3] In regards to behavioural science it is known that people change their behaviour when they know they are being watched, also known as *"Hawthorne Effect"*.[18] Building on this premise, this research proposes an **unprecedented** and **novel** approach to hand hygiene promotion and education. The physical presence of a robot could impact human perception and influence behaviour as humans have a tendency to anthropomorphize social robots.[19] This research will extend the scope of social robotics research by **filling the gap** between our understanding of influence and perception of social robots on handwashing behaviour.

- **Aim:** To develop **persuasive strategies** for social robot to influence handwashing behaviour and investigate its impact by systematic empirical studies in schools. The key research questions to be addressed in this research are **(i)** what interaction design strategies (verbal and non-verbal) by the social robot are the most effective in influencing handwashing behaviour? and **(ii)** how can social robots achieve sustainable handwashing behaviour change?

- **Objectives:** **(i)** Develop an interdisciplinary approach combining behavioural science, gamification principles and multimodal affective feedback strategies from the social robot; **(ii)** Implement a user-centered iterative design approach within the target population; **(iii)** Perform empirical studies in rural schools to investigate the influence of a social robot on handwashing behaviour.

- **Approach:** The approach will focus on investigating the use of gamification which has shown positive results in technology based education[20] and health related contexts.[21] Gamification is the process of including game like elements that could motivate and influence human behaviour to enhance learning. However, gamification strategies have not been investigated before in the context of handwashing for subjects from rural backgrounds in developing countries using social robots. My previous work with affective feedback has shown promising results with social robots as tutors.[22] I will investigate whether both extrinsic and intrinsic motivation to wash hands can be influenced by gamification combined with affective feedback strategies.

Perception: The technical challenge is to recognise hand gestures and behaviour cues in outdoor in the light sensitive and unstructured environment due to diverse physical setups of handwashing stations in different schools. To determine the quality of handwashing, motion detection using background subtraction algorithm within a region tightly cropped area at the water tap will be developed (fig. 2). Using the examples of hand washing video gathered from user studies, a computer vision algorithm will be developed using deep learning methods (CNN and LSTM).[23]

Synthesis: For affective expressiveness, the social robot will also have eye movement to track users and display emotions e.g. happy, sadness, disgust. A screen to display graphical mouth to display smiles, frowns and hand hygiene educational videos and a loud speaker for sounds and speech synthesis in the local language. Gamification strategies can include for example, if the children wash their hands effectively, a time based graphics will be shown on the screen (e.g. balloon bursts or a rocket fires). Verbal and non-verbal feedback strategies using social reinforcement[16] will also be used (e.g., the robot expressing social praise by saying "Great, keep it up" while showing a happy facial expression and sounds). The perception and synthesis modules will be combined into an overall system architecture (fig. 2).

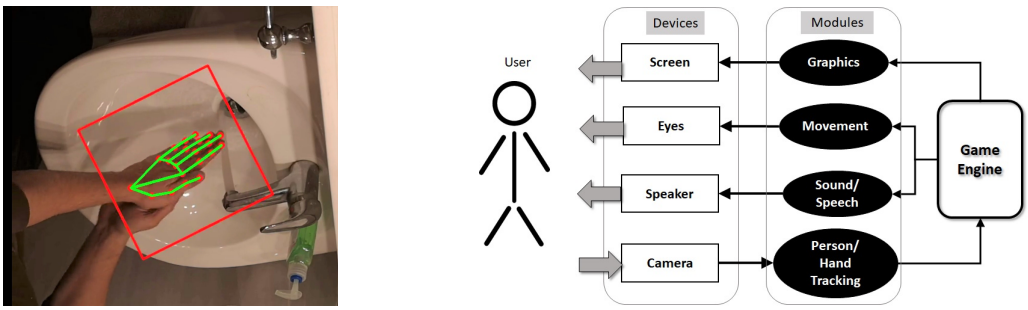


Figure 2: Image Left: Tracking hand movement (Test Video), Image Right: System Architecture

Analysis: To achieve the objectives mentioned above (i, ii, iii), mixed research methods will be used i.e. qualitative and quantitative data will be collected analysing the videos and questionnaires from user studies. **(1)** The influence of the social robot will be analysed by measuring verbal and non-verbal behaviours that positively correlate with desirable action from the participant, e.g. responding to the instructions from the robot and their reaction time. Social signals from the users such as proxemics (i.e. coming close to the robot), eye contact (frequency, duration), speech utterances, and body movements (gestures, head pose) to measure the degree of engagement with the robot. **(2)** Learning transfer will be evaluated by analysing the handwashing behaviour i.e. quality of handwashing will be measured by how many steps the children carried out of the 6 steps of handwashing[24] and the duration and frequency of handwashing during critical moments such as before meals and after use of toilet. **(3)** The questionnaires will capture perceptions about appearance, verbal and non-verbal behaviours of the robot.

• **The Fellowship as a Springboard to Success over the next 10 years and beyond:** My research vision is to develop low-cost high impact social robots as agents of positive behaviour change for people in developing countries. The proposed research will provide me with a platform on which to build well beyond the 5 year term of the Fellowship. The outcomes of the proposed research will contribute towards informing the design of future social robots, both for the challenges of the developing world and also broader applications for effecting positive behaviour change. During the fellowship, having connected to rural communities, other use-cases for social robots will be investigated via discussions with stakeholders. The future research areas will be centred around UN's sustainable goals on topics such as, WASH (Water Sanitation and Hygiene), Good Health Well-being, Quality Education, Addiction Awareness, and Gender Equality. I will apply for research grants with external collaborators in the area of technology based interventions for positive behaviour change in developing countries. After the fellowship I see myself leading a research group to advance social robotics research for developing countries.

References

[1] P. Baxter et al. In: *2016 11th ACM/IEEE International Conference on Human-Robot Interaction (HRI)*. IEEE. 2016, pp. 391–398. [2] D. T. Jamison et al. The World Bank, 2006. [3] E. De Buck et al. In: *Campbell Systematic Reviews* 7 (2017), pp. 1–447. [4] T. Belpaeme et al. In: *Science robotics* 3.21 (2018), eaat5954. [5] G. Gordon et al. In: *Thirtieth AAAI Conference on Artificial Intelligence*. 2016. [6] R. Dreibelbis et al. In: *International journal of environmental research and public health* (2016). [7] A. Deshmukh et al. In: *RO-MAN*. IEEE. 2019. [8] A. Biran et al. In: *The Lancet Global Health* 2.3 (2014), e145–e154. [9] V. Curtis et al. In: *The Lancet infectious diseases* 11.4 (2011), pp. 312–321. [10] G. Judah et al. In: *American Journal of Public Health* 99.S2 (2009), S405–S411. [11] S. Storey et al. In: *Journal of Hospital Infection* 88.2 (2014), pp. 84–88. [12] D. A. Fisher et al. In: *Infection Control & Hospital Epidemiology* 34.9 (2013), pp. 919–928. [13] A. K. Venkatesh et al. In: *American journal of infection control* 36.3 (2008), pp. 199–205. [14] S. Gaube et al. In: *PloS one* 13.5 (2018). [15] A. Haque et al. In: *Proceedings of Machine Learning for Healthcare* (2017). [16] J. Ham and C. J. Midden. In: *ICSR* 6.2 (2014), pp. 163–171. [17] K. Shinozawa et al. In: *IROS*. Vol. 2. IEEE. 2004, pp. 1908–1913. [18] H. A. Landsberger. In: 3.2 (1960). [19] C. Breazeal. In: *Annual international conference of the IEEE engineering in medicine and biology society*. 2011, pp. 5368–5371. [20] S. Kiesler et al. In: *Academic exchange quarterly* 15.2 (2011), pp. 1–5. [21] P. Pereira et al. In: *International conference of design, user experience, and usability*. Springer. 2014, pp. 742–753. [22] M. E. Foster et al. In: *AAMAS*. 2015, pp. 1711–1712. [23] G. Zhu et al. In: *Ieee Access* 5 (2017), pp. 4517–4524. [24] https://www.who.int/gpsc/5may/How_To_HandRub_Poster.pdf. 2009.

Impact Statement

• **Academic and public Impact:** Meta analysis of the results from all field studies will be consolidated into journal publications for example International Journal for Social robotics, Science direct, Computers & Education and **high REF impact** Open Access journals such as Lancet Global Health. The main outcome from the meta analysis will provide **new understanding** how people rationalise their behaviour and be useful in designing appropriate interaction strategies using social robots in broader scenarios as agents for **positive** behaviour change. Publications will be written throughout the fellowship. The results will be further disseminated through social media channels, press releases, local meetings, workshops in rural schools in India also in the UK. Public engagement activities, Seminars and events such as Global Handwashing Day and science festivals. I will engage with NGOs and other stakeholders for example, public-private partnership with companies like Unilever (Lifebuoy) who are involved in handwashing programmes worldwide.[10] I will seek to diversify to other countries through University of Manchester's international network.

• **Economic Impact:** With my proposed research I initially target India, which has the world's largest rural population of 850 million, with only 18% practising handwashing with soap. The annual net costs to India from not washing hands with soap are estimated at US **$ 23 billion** and US **$ 8.955 billion** for children aged between 5–14 years[2]. Since the target age group for the proposed research is primary school children aged between 5-11. A national handwashing program in India could provide an estimated **92-fold return** on investment. We are not targeting the entire population in our research, but the potential cost savings from this research if put to scale can produce huge economic savings. Also relevant to Europe, in the UK with 32,000 schools, a survey by the London School of Hygiene & Tropical Medicine showed that only 63.8 % of primary school and 39% for secondary school children regularly washed their hands with soap. The total annual cost for treating children with infections in primary care was estimated to be **£757 million** in the UK. A hand hygiene intervention could bring estimated savings ranging from £106 to £387 per child per year in the UK[3]. There is also a potential to apply this research for school children in the UK.

• **Societal Impact:** Handwashing with soap is included in the United Nations (UN) sustainable development goals Target 6.2 for sanitation and hygiene. Every year, nearly 443 million school days are lost due to water and sanitation related diseases in developing countries. My proposed approach has a potential to increase hand hygiene compliance, which in return could **reduce** germ transmission, save public health costs and decrease infection-related absenteeism in schools. It will also **contribute** towards meeting UN's sustainable development goals targets around sanitation, hygiene, education, nutrition and child survival in developing countries. This research aims to go above a purely research-driven agenda and with a focus on high impact on society which also aligns with the EU "Science with and for society" initiative[4]. This research can bring economic, social and health benefits to communities. The UN's sustainable development goals also relate to bridging the digital divide, making technology more accessible to all users rather than a privileged few in developed countries. This project will also help to promote social inclusion and help bridge the digital divide.

• **Impact on policy makers:** This project can have a major impact on global health policy makers. An email response from Scottish Government Minister, Lara Allan Chief nursing officer, dated 16th March 2020 emphasised "The social robot looks like an interesting and creative concept to encourage effective handwashing by children, which is particularly important in the current climate." Given the current context with COVID-19 the proposed research has a far greater potential impact globally [5]. Social robots could have a great potential to serve a supportive role in large scale health interventions both as active agents of behaviour change as well as objective assessors of various indicators of health and hygiene.

[2] Townsend et al., (https://doi.org/10.1111/tmi.12808,2017)

[3] https://www.birmingham.ac.uk/Documents/college-mds/haps/projects/WMHTAC/REPreports/2006/Handwashing.pdf

[4] https://ec.europa.eu/programmes/horizon2020/en/h2020-section/science-and-society

[5] News about my research covered on BBC news, see around 3.20 minutes in the video https://www.bbc.co.uk/news/av/technology-52105478/how-robots-have-joined-the-battle-against-coronavirus